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Faculty of Engineering

CHEP – Communications Systems Engineering Program

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Sheet 4

Spatial Filtering

Section 1: 1-D Spatial Filtering

Problem 1. (1-D Filtering — Average & Median)

(a) Consider a row of pixels with values:

$$1,\ 1,\ 1,\ 1,\ \mathbf{5},\ 1,\ 1,\ 1,\ 1$$

Apply a **size-3 average filter** and a **size-3 median filter** to this row. What are the output values at the **5th pixel** (the outlier) for each filter?

(b) Consider a row of pixels with values:

$$1,\ 1,\ 1,\ 1,\ \mathbf{5},\ 5,\ 5,\ 5,\ 5$$

Apply a **size-3 average filter** and a **size-3 median filter**. What are the output values at the **5th pixel** for each filter?

Hint:

- Average filter: $\hat{p} = \frac{1}{3}(p_{i-1} + p_i + p_{i+1})$
- Median filter: $\hat{p} = \text{median}\{p_{i-1},\ p_i,\ p_{i+1}\}$

Problem 2. (1-D Signal: Identify Mean vs. Median Filter)

The top signal below is the original 1-D signal. The two signals below it are the results of applying either a **mean filter** or a **median filter** (in some order) to the original.

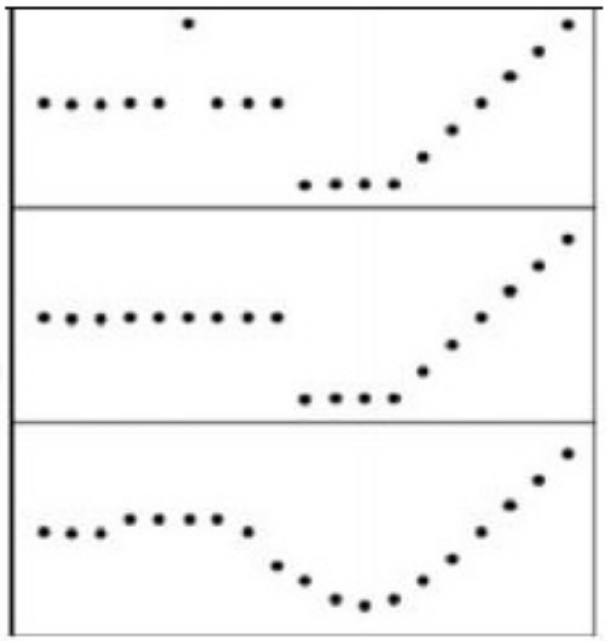

Figure 1: Original signal (top); filtered signal 1 (middle); filtered signal 2 (bottom).

State which filtered signal corresponds to the mean filter and which to the median filter. Give clear reasoning for each assignment based on the expected behaviour of the two filters.

Section 2: 2-D Spatial Filtering

Reference Image (Used in Problems 3–6):

The 5×5 image below is indexed by column and row as $[x,\, y]^T$ starting at $[0, 0]^T$ (top-left corner). The value 48 at $[1, 2]^T$ is an **outlier pixel**. Use **zero-padding** at boundaries unless stated otherwise.

Problem 3. (3×3 Average & Median Filter)

(a) Using the averaging kernel

$$H = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

compute the output intensity at pixels $[2, 0]^T$ and $[3, 2]^T$.

100	102	104	102	106
102	105	107	106	106
104	**48**	108	109	110
105	110	104	106	109
107	105	103	107	105

Table 1: Reference 5×5 image. Bold value = 48 is the outlier at $[1, 2]^T$.

(b) Apply a 3×3 median filter and compute the output intensity at pixel $[2, 1]^T$.

Problem 4. (Gaussian Filter)

Apply the Gaussian kernel

$$H = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

and compute the output intensity at pixel $[2, 2]^T$.

(*Note: the kernel weights sum to 16.*)

Problem 5. (Reverse-Engineering a Linear Filter)

After passing the reference image through an unknown linear filter, the following output is obtained:

0	0	0	0	0
0	100	102	104	102
0	102	105	107	106
0	104	48	108	109
0	105	110	104	106

Table 2: Output image after the unknown filter.

(a) Identify the filter matrix (kernel) that produced this output.
(b) State how boundary pixels were handled.
(c) Justify your answer by verifying at least one output pixel.

Problem 6. (Blurring followed by Median on a Binary Image)

Image $b(x, y)$ is binary with equal numbers of 0 and 255 pixels. Two filters are applied **sequentially**:

- A 3×3 blurring (cross-shaped) filter:

$$H = \begin{bmatrix} 0 & \frac{1}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & 0 \end{bmatrix}$$

- A 3×3 median filter.

(a) If the blurring filter is applied **before** the median filter, what are the **possible output pixel values** of $o(x, y)$? Justify your answer.
(b) Construct a binary image $b(x, y)$ for which the output $o(x, y)$ is the **same regardless of filter order**.

Section 3: Basic Filter Computations

Problem 7. (Computing Filter Output)

Given a 3×3 image region and an averaging filter:

Image Region:

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Filter:

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find the output value at the center pixel.

Problem 8. (Median Filter Computation)

Given a 3×3 neighborhood with values:

$$\begin{bmatrix} 100 & 10 & 120 \\ 15 & 200 & 18 \\ 110 & 12 & 115 \end{bmatrix}$$

(a) Sort the values.
(b) Determine the median.
(c) Compare with the averaging filter result.

Note: The center pixel (200) appears corrupted by impulse noise.

Problem 9. (Laplacian Sharpening)

Given a 3×3 image region:

$$\begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & 5 \\ 5 & 5 & 5 \end{bmatrix}$$

(a) Apply the basic Laplacian mask (center $= -4$).
(b) Apply the Laplacian with diagonals (center $= -8$).
(c) Compute the sharpened output: $g(x, y) = f(x, y) - \nabla^2 f$.

Problem 10. (Sobel Gradient Magnitude)

Given a 3×3 image region:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Use the Sobel operators:

$$g_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad g_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Find the gradient magnitude $M(x, y) = |g_x| + |g_y|$ at the center pixel.

Problem 11. (Unsharp Masking)

Given:

- Original pixel value: $f(x, y) = 100$
- Blurred value: $\bar{f}(x, y) = 80$

Compute the sharpened output using:

$$g_{\text{mask}}(x, y) = f(x, y) - \bar{f}(x, y)$$

$$g(x, y) = f(x, y) + k \cdot g_{\text{mask}}(x, y)$$

(a) Unsharp masking with $k = 1$.
(b) Highboost filtering with $k = 2.5$.

Problem 12. (Gaussian Kernel Size)

Given a standard deviation $\sigma = 5$, determine:

(a) The minimum kernel size using: size $\geq 6\sigma$.

(b) What is the smallest odd kernel size?
(c) Why do we use odd-sized kernels?
(d) What happens if we use a larger kernel (e.g., 85×85)?

Concept: The 6σ rule ensures we capture 99.7% of the Gaussian distribution.

Section 4: Correlation vs. Convolution

Problem 13. (Correlation vs. Convolution)

Given a filter kernel:

$$w = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

(a) Perform correlation with an image region.
(b) Perform convolution (rotate the filter 180°).
(c) Compare the two results.
(d) When are correlation and convolution equivalent?

Section 5: First vs. Second Derivatives

Problem 14. (First vs. Second Derivatives)

Given an intensity profile:

$$[5,\ 5,\ 5,\ 8,\ 12,\ 15,\ 15,\ 15]$$

Compute:

(a) First derivative: $\dfrac{\partial f}{\partial x} = f(x+1) - f(x)$
(b) Second derivative: $\dfrac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$

Analyze the response in:

- Constant regions
- Start of ramp
- Along the ramp
- End of ramp

Section 6: Image Padding Effects

Problem 15. (Image Padding Effects)

Given a 5×5 image, apply a 3×3 averaging filter.

Compare three padding methods:

(1) **Zero padding:** Add zeros around the border.
(2) **Replicate padding:** Repeat edge pixels.
(3) **Mirror padding:** Reflect the image across the border.

(a) Compute the filtered output for each method.
(b) Observe border effects.
(c) Which method produces the best results and why?

Section 7: Combined Enhancement Operations

Problem 16. (Combined Enhancement Operations)

Scenario: Enhance a bone scan image using a combination of techniques.

Steps:

0 Apply Laplacian: $f_L = \nabla^2 f$.
0 Sharpen: $f_1 = f + c \cdot f_L$.
0 Compute Sobel gradient: $f_2 = \text{sobel}(f)$.
0 Smooth the gradient: $f_3 = \text{blur}(f_2)$.
0 Create a mask: $f_4 = f_1 \times f_3$.
0 Final result: $g = f + f_4$.

Derive the final equation $g(x, y)$ in terms of the original $f(x, y)$.

Section 8: Textbook Problems

Problem 17. [1] **(Order of Operations: Averaging then Laplacian)**

In an application, an averaging mask is first applied to reduce noise, and then a Laplacian mask is applied to enhance fine details. Would the result be the same if the order were reversed? Justify your answer using the properties of linear operations.

(*Recall: Both averaging and the Laplacian are linear spatial filters. For linear filters,* $F_1 * (F_2 * I) = F_2 * (F_1 * I)$ *since convolution is associative and commutative.*)

[1] R.C. Gonzalez & R.E. Woods, *Digital Image Processing*, 4th ed. (Pearson, 2018), Problem 3.23.

Problem 18. [2] **(Averaging Mask Size Effects)**

Three images were blurred with square averaging masks of sizes $n = 23$, $n = 25$, and $n = 45$ respectively. The vertical bars in images (a) and (c) are blurred but clearly separated, while the bars in image (b) have merged—despite image (b) using a *smaller* mask than image (c).

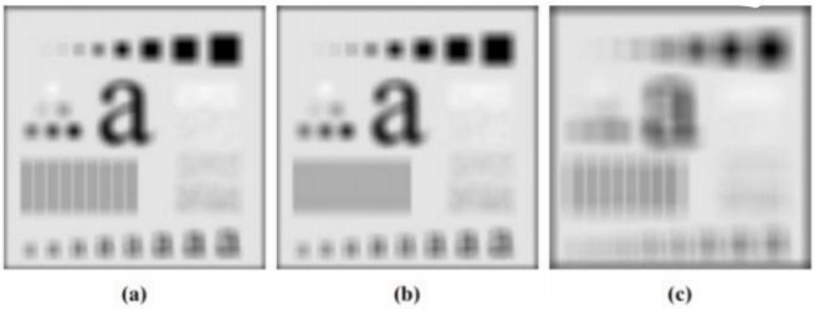

Figure 2: (a) $n = 23$: bars separated; (b) $n = 25$: bars merged; (c) $n = 45$: bars separated.

Explain why the bars in image (b) merged while those in image (c) did not, even though image (c) used a larger mask.

[2] Gonzalez & Woods, 4th ed., Problem 3.21.

Problem 19. (Identifying the Smoothing Filter Used)

The left image was filtered to produce the right image. The small black square in the lower-right corner of the original indicates the mask size used.

Figure 3: Original image (left) and filtered output (right). The black square shows the mask size.

The filter used was **one** of the following:

(1) Averaging (box) filter
(2) Ideal low-pass filter
(3) Gaussian low-pass filter
(4) Median filter

For **each** of the four filters, give at least one reason why it **was** or **was not** the filter used to produce the output shown.

Problem 20. [3] **(Same Histogram, Different Images After Blurring)**

The two binary images shown below are visually quite different but share the same histogram. Each image is blurred with a 3×3 averaging kernel.

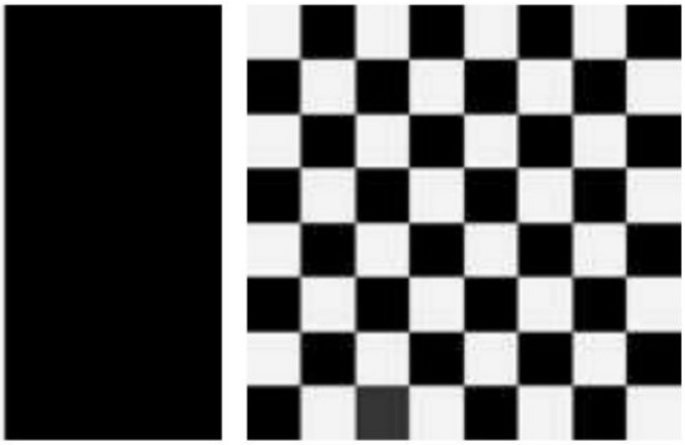

Figure 4: Left: half-black image; Right: checkerboard image.

(a) Would the histograms of the two blurred images still be equal? Explain.
(b) If not, sketch the two resulting histograms and describe how they differ.

[3] Gonzalez & Woods, 4th ed., Problem 3.14.